STAT 580 - Time Series

STAT 471: Introduction to R Programming Lecture

Lecture 14: Introduction to Time Series

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Overview

1. What is a Time Series?
2. Basic Analyses of Time Series
3. More Advanced Techniques: Change Point and Anomaly Detection

mean and
variance changes

Outliers

Revisiting Stochastic Processes

Before introducing the formal time series definition, we can revisit the idea of stochastic processes: Stochastic processes can be useful to represent random variables and their evolution over time or space.

Definition: Stochastic Process

A stochastic process $\{X(t) : t \in T\}$ is a family of random variables where t denotes the time step and T is a nonempty set. A stochastic process is discrete if T is finite, otherwise if T is infinite, then it is continuous.

- We've seen a few examples of these: random walks, autoregressive random walk, moving average, and white noise
- Time series are not that different!

$WN(0, \sigma^2)$

t = time step, lag

Time Series Formal Definition

stocks, financial analysis, epidemiology

Definition: Time Series

A time series $\{Y(t) : t \in T\}$ is a collection of random variables where t denotes the time step and T is a nonempty set. and are indexed according to the order they are obtained in time. Time series models can help in modeling stochastic processes and their fluctuations in time.

Definition: Stationary Time Series

A time series $\{Y(t) : t \in T\}$ is (weakly) stationary if the following two conditions hold: (a) The mean of the time series is constant and does not depend on time t and (b) the autocovariance function $\gamma(s, t)$ depends only on s and t through their difference $|s - t|$. In short, a time series is stationary if its distribution does not depend on t (not saying the same thing as (b)).

Stationarity Check Example

Is the time series process $Y_t = \epsilon_t + \beta_1 \epsilon_{t-1}$ where $\epsilon_t \sim WN(0, \sigma^2)$ stationary?

Stationary Check Example

① $E[Y_t] = \underbrace{E[\varepsilon_t]}_{0} + \beta_1 \underbrace{E[\varepsilon_{t-1}]}_{=0} = 0$ so there is no t

② $\boxed{\gamma(s,t)} = E[Y_t Y_{t-\tau}]$ $\leftarrow$ $\tau = \underbrace{1, 2, 3, \ldots}$

$= E[\varepsilon_t + \beta_1 \varepsilon_{t-1}] E[\varepsilon_{t-\tau} + \beta_1 \varepsilon_{t-1-\tau}]$

$= E[\varepsilon_t \varepsilon_{t-\tau}] + \beta_1 E[\varepsilon_t \varepsilon_{t-1-\tau}] + \beta_1 E[\varepsilon_{t-1} \varepsilon_{t-1-\tau}]$
$+ \beta_1^2 E[\varepsilon_{t-1} \varepsilon_{t-\tau-1}]$

$= \beta_1 E[\varepsilon_{t-1} \varepsilon_{t-\tau}] = \begin{cases} \beta_1 \sigma^2 & \text{if } \tau = 1 \\ 0 & \text{if } \tau > 1 \end{cases}$ Does not depend on t

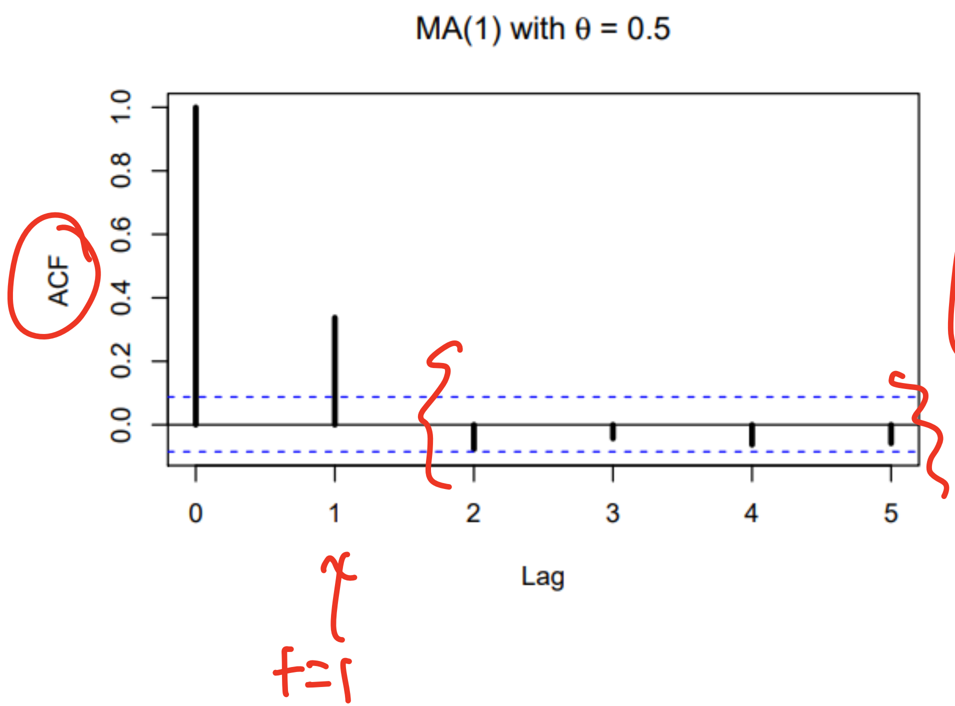
Y_t is stationary.

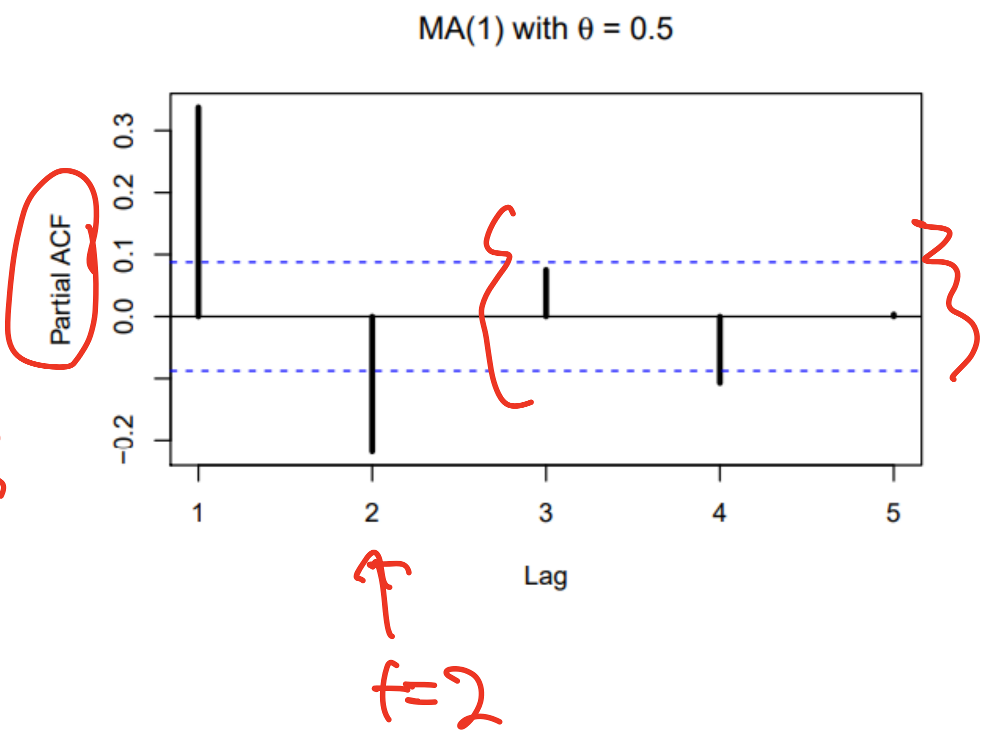
Checking for Stationarity: Auto~~covariance~~ correlation Functions and Partial Auto~~covariance~~ correlation Functions (ACF and PACF)

autocorrelation autocorrelation

We can plot ~~autocovariance~~ and partial ~~autocovariance~~ functions to detect stationarity visually. In an ACF plot, we want to see the tick marks (the autocovariances) go down to zero or below the horizontal dashed lines.

Example


MA(1) with θ = 0.5
ACF
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
Lag
t=1


MA(1) with θ = 0.5
Partial ACF
0.3
0.2
0.1
0.0
−0.2
1
2
3
4
5
Lag
t=2

Augmented Dickey-Fuller (ADF) and the KPSS Hypothesis Tests

Check: p-value < 0.05

These hypothesis tests can be used to diagnose stationarity in time series data.

ADF

H_0 : Time series is non-stationary
H_a : Time series is stationary

KPSS

H_0 : Time series is stationary
H_a : Time series is non-stationary

We can use p-values to determine our conclusions at a deterministic significance level.

MA(q) ← q is the order for a MA time series

We will now consider a general finite-order moving average process of order q, MA(q):

$$Y_t = \underbrace{\epsilon_t + \theta_1 \epsilon_{t-1} + \ldots + \theta_q \epsilon_{t-q}}_{} = \Theta(L)\epsilon_t, \quad -\infty < \theta < \infty, \quad \underbrace{\epsilon \sim WN(0, \sigma^2)}_{}$$

where

$$\boxed{\Theta(L) = 1 + \theta_1 L + \ldots + \theta_q L^q}$$

is the qth-order lag polynomial. The MA(q) process is a generalization of the MA(1) process. Compared to MA(1), MA(q) can capture richer dynamic patterns which can be used for improved forecasting.

MA(2): $\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$

$= \varepsilon_t + 2\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$

AR(p)

The general *p*th order autoregressive process, AR(p) is:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + ... + \phi_p Y_{t-p} + \epsilon_t, \quad \epsilon_t \sim WN(0, \sigma^2)$$

In lag operator form, we write:

$$\boxed{\Phi(L) Y_t = (1 - \phi_1 L - \phi_2 L^2 - ... - \phi_p L^p) Y_t = \epsilon_t}$$

AR(1): $\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_2 Y_{t-2} + \varepsilon_t$

$\phi_1 Y_{t-1} + 2\phi_2 Y_{t-2} + \varepsilon_t$

Summary of Some Time Series Models: Auto-Regressive Moving Average

ARMA(p,q)

AR(p) + MA(q)

A natural generalization of the ARMA(1,1) is the ARMA(p,q) process that allows for multiple moving-average and autoregressive lags. We can write it as:

$$Y_t = \phi_1 Y_{t-1} + \ldots + \phi_p Y_{t-p} + \epsilon_t + \theta_q \epsilon_{t-1} + \ldots + \theta_q \epsilon_{t-q}, \quad \epsilon_t \sim WN(0, \sigma^2)$$

or:

$$\Phi(L) Y_t = \Theta(L) \epsilon_t$$

AR

MA

ARIMA(p,d,q)

If we combine differencing with autoregression and a moving average model, we obtain a non-seasonal ARIMA model. ARIMA is an acronym for AutoRegressive Integrated Moving Average (in this context, "integration" is the reverse of differencing). The full model can be written as

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t, \qquad (8.1)$$

$\varepsilon_t \sim N(0, \sigma^2)$

d = 1 or 2 makes your time series stationary

where y'_t is the differenced series (it may have been differenced more than once). The "predictors" on the right hand side include both lagged values of y_t and lagged errors. We call this an **ARIMA(p, d, q) model**, where

$p =$	order of the autoregressive part;
$d =$	degree of first differencing involved;
$q =$	order of the moving average part.

d = 1st difference

The same stationarity and invertibility conditions that are used for autoregressive and moving average models also apply to an ARIMA model.

$$y'_{t(1)} = c + \phi_1 y'_{t-2} + \dots + \phi_p y'_{t-p-1} + \theta_1 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q-1} + \varepsilon_t$$

Summary of Some Time Series Models: Seasonal Auto-Regressive Integrated Moving Average

SARIMA(p,d,q)(P,D,Q)[m] seasonal parameters

AirPassengers had a nonlinear trend

non-seasonal part of time series

So far, we have restricted our attention to non-seasonal data and non-seasonal ARIMA models. However, ARIMA models are also capable of modelling a wide range of seasonal data.

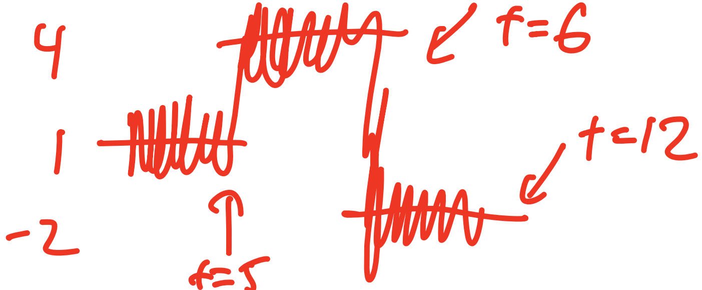
A seasonal ARIMA model is formed by including additional seasonal terms in the ARIMA models we have seen so far. It is written as follows:

$$\text{ARIMA} \quad \underbrace{(p,d,q)}_{\substack{\uparrow \\ \text{Non-seasonal part} \\ \text{of the model}}} \quad \underbrace{(P,D,Q)_m}_{\substack{\uparrow \\ \text{Seasonal part of} \\ \text{of the model}}}$$

where $m =$ number of observations per year. We use uppercase notation for the seasonal parts of the model, and lowercase notation for the non-seasonal parts of the model.

If time series is annual, ⇒ $m=12$ ← seasonality

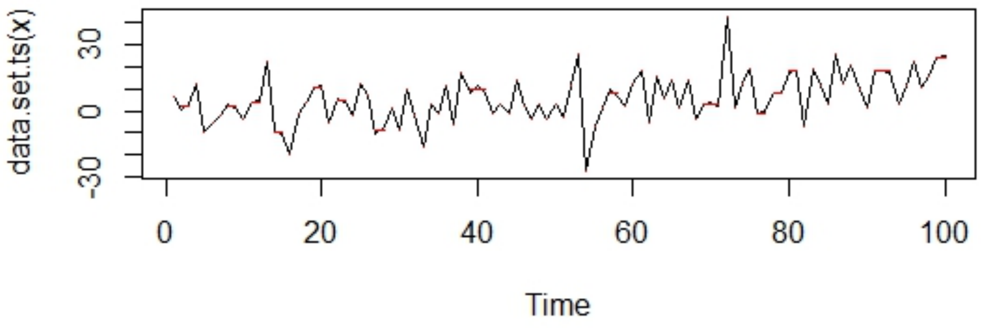
Change Point Detection

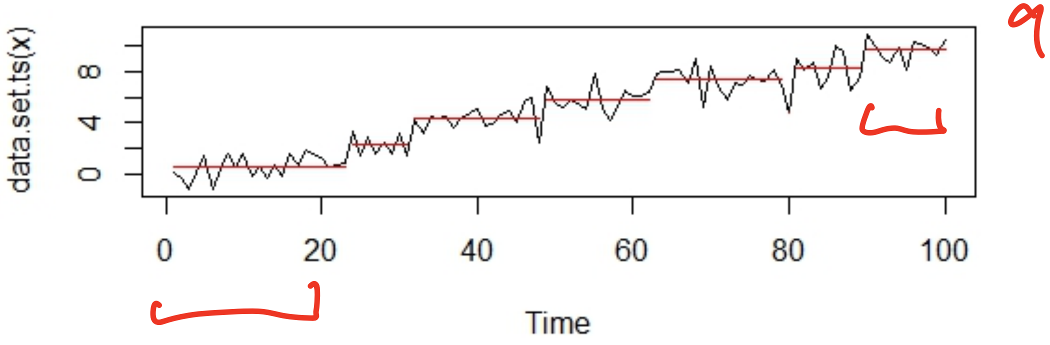


A change point is a specific set of time points in a time series where the mean or variance changes. While this can be done by-hand (not the best method), we can use the Pruned Exact Linear Time (PELT) algorithm to detect multiple change points for means, variances, or both. The PELT method is efficient since the runtime is linear. We will use the changepoint package in R to carry out the algorithm.

Optimal runtime

Change Point Detection Example

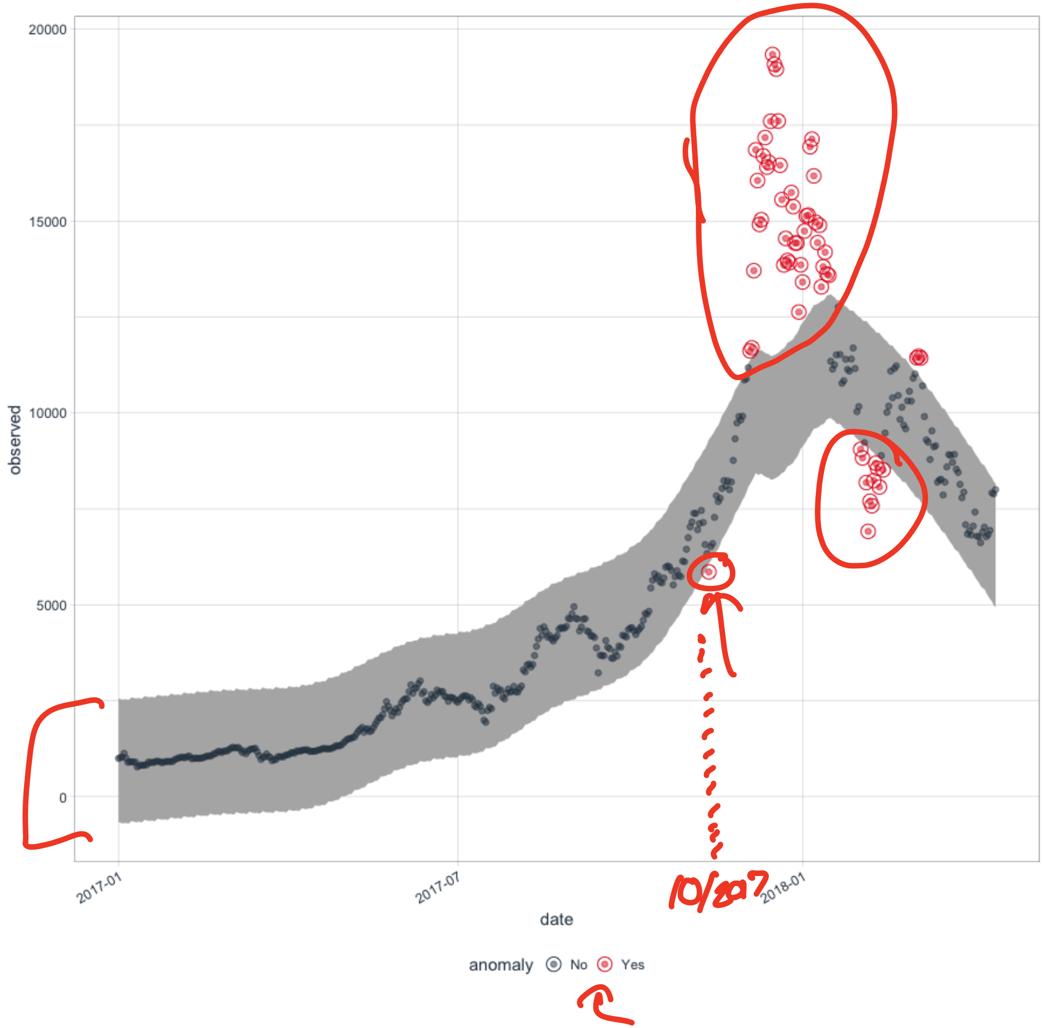




Anomaly Detection

The anomalize package in R provides a Tidyverse approach to analyzing the time series and detecting outliers (anomalies). We first decompose the time series, then use anomalize() which uses either the IQR method or the Generalized Extreme Studentized Deviate Test (GESD) to detect the anomalies and recomposes the time series with the points labeled.

Anomaly Detection Example



Next Time...

Your Turn! You'll be presenting your projects during finals week.

Announcements

Homework 6 due this Saturday at 11:59pm!